

RND-6: SAFE IN-TIME COVARIATES ??? CUPAC-STYLE ????????

> ??????: ? ??????. ????????????? ?????????????? ????????? ? ??????? ???????
> A/B ? CUPAC A ? A+B ? A+B+C; ?????????? ?????????? ?????????? open-source ?????????? ?
> ?????????? ?????????????? ?????????? ? delta-???????????? ?? A-only (??????? «Expanded public dataset
> validation»). ?????????? ??????? ?? B/C-???????? ? notebook.ipynb (???????????, seed=11).

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???????????, ??? ? ? ?????????????????? ??????????? ??????????? in-time ?????????? ??????
scikit-CUPAC ?????????????????? ?????????? ?????????? (? ??????????? ?????????? ??????????) ???
?????????? ?????????????????? ?????????????? ??????? ATE.

?????? ?????????? (SAFETY LAYER)

? **A** pre-treatment ? ??? ? CUPAC;
? **B** expert-safe in-time ? ?????????? second-stage ??????????;
? **C** balance-gated in-time ? ?????????? ?????????? ????? balance/missingness gate;
? **D** dag-required ? ??? estimator (????? causal-????????);
? **E** mediator-risk, **F** leakage ? ?????????? ??? ?????????? ?????? ATE;
? **unsafe_demo** ? ?????? ?????????????? ??????, ?? ??????????

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ab_hypex (A/B baseline) • hypex_cupac (reference/parity, ??? ??????) •
sklearn_cupac_A ? sklearn_cupac_A_plus_B_linear ? sklearn_cupac_A_plus_B_plus_C_linear
(????????? ?????????? ??????????) • unsafe_demo_optional (?????????????).

?????????? ? ? ?????????? (N=8000, SEED=11, ?????????? ATE = 2.8)

method	ate	p_value	variance_reduction_vs_ab_%	incremental_vs_predecessor_%	safety_status
ab_hypex	2.758	0.0	0.0	?	ok
hypex_cupac	2.785	0.0	13.56	?	reference_only
sklearn_cupac_A	2.801	0.0	25.63	25.63	ok
sklearn_cupac_A_plus_B_linear	2.775	0.0	42.38	22.52	ok
sklearn_cupac_A_plus_B_plus_C_linear	2.766	0.0	52.80	18.07	ok
unsafe_demo_optional	1.000	0.0	83.17	?	unsafe_demo

(????????? ?????????? ?????????? ? ?????? ?????????????? ?????????? ?? ?????????? ??????????)

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- ?????? ?????????????? ??????? ???? ?????????????????? ?????????: A (~26%) ? +B (~42%) ? +B+C (~53%)
????????? ?????????? ?????????????? A/B, ??? ??? ATE ?????????? ?????????????? (? 2.8).
- CUPAC parity: hypex_cupac (~14%) ?????????? ?? ?????? ? sklearn_cupac_A;
????????? ?????????? ??????? ??? R&D ? ?????????????? ??? ??????????
- Balance/missingness gate ?????????? A/B/C (|SMD|>0) ? ?????????? D/E/F ? unsafe_demo
(|SMD| ? 0.45?1.14, p>0) ? ?????????, ?????? ?? ?????? ?????? ? estimator.

4. unsafe_demo ???? ???????????? «????????? ??????????» (~83%), ?? ??????? ?????? ATE
?????? ?? ????????? 2.8 ? ????????? ? 1.0 (?. ??????) ? ????????????? ??????
outcome-derived ??????????; ? ????????????? ?????? ?? ??????.

????????????? ? ?????????????

? balance-test ? ????????????? gate (????? ????????????? ?????????? E/F), ? **??** ??????????????
????????? ?????????????; ??? D-????????? ?????? ??????? DAG/causal-?????
? E/F ??????????; safe-intime ????????????? **????** ? ????????? second-stage, ?? ?
???????????? ML-?????.

?????? ?????????? ?? ????????? ??????

? **????????** ????????????? ?????? ??????? **A+B+C** (??????? ????): ?????? ??? ??????
A/B/C/D/E/F ?????????? ?? ??????????
? **????????? ?????????? ?????????? ?????? ?????????????? ? ????????? A-only** CUPAC-?????????
????? ????? (download + schema_summary) ?????? Orange Belgium, Lenta, Criteo, X5
RetailHero,
MegaFon, Lazada/DESCN, CUVET-policy, MT-LIFT, Dunnhumby ? semi-synthetic causal-?????????
? **Hillstrom** ?????????? pipeline/A-only, **??** B/C.
? ????????? safe in-time **B/C** ?????????????: ?????????? (f*, 'PC*', 'X_*'),
?????
??? ???-????????? ?????????? (A), ??? ????-treatment ?????????? (E/F). ?????????? ? ??????????
????????????????? ??? ???? ?????? (X5), ??? ? ???? ????????????? (Dunnhumby).
? **????? A+B+C-????????? ? ?????????? ?????????? ?????????? ? ?????????????** ? ?????????
????????????????, ??? ???? ???? ? ???? ???? B/C-????? ? timing-evidence.
? ?????? ? ????????????? ?????????????? ??????????: **docs/06_real_dataset_raw_audit.md**.

EXPANDED PUBLIC DATASET VALIDATION

????????????? ?????????? (????? ??? ???? ????????????? uplift-????? + ?????????? ?????????/????????? +
????????? ?????????? + ?????????????? delta-????? ??????). ?????? ?????? ? ??????
?????????????????????: docs/06_expanded_dataset_scouting.md.

????????????????????? ?????????? ?????????? (?? ?????? ???, ?? ?? ??????????????):
? **A** (pre-treatment) ? **9+** ?????????? ?????????? ?????? A-????????? ?? ?????? (+
?????????), ??
????????????? delta-????? ?????????? ? ?????????????? ?? 5 ?? ??? (hillstrom, orange_belgium,
lazada_descn, lenta, x5 ? ?????? ???). ?????????? A/sandbox-????????? ?????????? ?? ??????
?????/???????????????? (criteo ? access_blocked; open_bandit/dunnhumby/criteo_private_ad ?
sandbox,
?? ?????????? ?? ?????????? A-?????????).
? **B** (expert-safe in-time) ? **0**?: ?? ??? ???? ?????????? ?????? ?? ?????????????? ?????????????,
????????????????? ?????????????????? in-time ??????????-????????????? ???????.
? **C-????????** (????????, ?????????) ? **0**; sandbox-????????????? C ?????????????? ??????????
?? 2 event-log ?????????? (Open Bandit, dunnhumby), ?? ??? ? ???? ?????? ??????????
? **D-????????** ? **5** (Criteo 'exposure'; Lazada biased-train propensity; Open Bandit
action/position/pscore; CriteoPrivateAd display/privacy-????????; dunnhumby
campaign/coupon).
? **E/F/unsafe-????????** ? **7+** (Hillstrom 'visit'/'conversion'; Criteo 'exposure';
X5 ???-????????????????? ?????????; Open Bandit reward-????????????; CriteoPrivateAd delayed
click/sale; dunnhumby redemption ? ??).

?????-A ???????? (real RCT/uplift): Hillstrom, Lenta, X5 RetailHero, Orange Belgium, Lazada/DESCN (???????????????? test), Criteo (????? A-only).
C/D-«????????» (?? A/B): Open Bandit (logged bandit), dunnhumby Complete Journey (observational), CriteoPrivateAd (advertising logs) ? ?????? ??? *????????????*
C/D-????????,
?? ?????????? causal-????????????? safe in-time.

> ?????? A+B+C-????????? ?? ?????????? ?????????? ?????????? ?? ??????????????. ??????
> C/D-????????????????????? ?????? ?????????????? ?? event-log / bandit / advertising / journey
> datasets, ?????????? ?? ?????????? ?????????? A/B uplift benchmark.

Delta-????????? ?? ?????????? ???????? (A-only ?????????? ab_hypex ? sklearn_cupac_A, seed=11).
????????? ? ???????? ?????????????? python tools/audit_datasets.py --delta; ?????????????? ?????????????? ?
expanded_dataset_delta_results.csv, ?????????? ? figures/.

dataset	validation_level	n	ATE (A/B)	var.reduction	CUPAC-A vs A/B, %	A-????????????	safety
hillstrom	A_only_real	64 000	0.597	0.06	18	ok	
orange_belgium	A_only_real	11 896	?0.0028	6.06	336	ok	
lazada_descn	A_only_real	181 669	0.0037	5.41	83	ok	
lenta	A_only_real	687 029	0.0075	14.36	196	ok	
x5_retailhero	A_only_real	200 039	0.0332	16.10	10	ok	

? **CUPAC-A ???? ?????????? ?????????? ?????????? 0?16%** ?? ?????????? ????????; ?????????? ?????? ???,
??? ?????? ?????????????????? pre-treatment ?????????? ? lenta (????????????? ???-????????? ??????????,
~14%) ? X5 (engineered ?????????? ??????????, ~16%). ATE ?????????? ?????????????? (????????????????).
? **hillstrom** ? pipeline/A-only smoke: ?????????? 'spend' ?????????? ?????????? A-????????? ??????
????????? (~0.06%).
? **????????? '+B' ? '+B+C' ?? ?????????? ?????????? ?? ??????????????????*: ?????????? B/C-????????? ???,
?????????
????????????????? delta-????????? B/C ??-????????? ?????????????????? ?????????? ?? ?????????????? (?????????
?????).
? **?? ?????? ? delta-?????????:** 'criteo' ? ?????????? ?????????? HTTP 403 (access_blocked);
open_bandit ? ?????? obp ?????????????????? ? pandas 2.x (DataFrame.drop("col", 1)),
runtime-blocked sandbox; dunnhumby / criteo_private_ad ? ?????????? ??????????????????
(research_sandbox/event_log_sandbox, demo-only) ? ?????????? ??????-based ResearchDataset ??
????????????? ?????????, ?? ??? observational/ad-????? (?? RCT) ? ? ?????????? A-only delta-?????????
??
????????? ? ??? safe B/C ?? ??????????????????.

????????????????? ?? ?????????? ? HYPEX

????????? ?????????????? ?????????????? ?????????? ?? ?????????????? ? ?????????????? ? HypEx CUPAC; ?? ??????????
????????????? ?????????? ?????????????????? A-only ?????? CUPAC-A (0?16% ?????????????? ??????????????, 5 ??????????????).
????????? ?????????????? ? HypEx ??????????????????:
1. ?????????????????? ?????? B/C ?? ?????????????????? ?????????????????? ?????????, ? ?? ?? ??????????????. ??????????????????
????????????? ??????????, ??? ?????????????? ?????????????? ?? ?????????????? B/C (????????????? ?????????????????? safe
in-time ?????????? ? timing-evidence ???; ??, «Expanded public dataset validation»). ??????????
B/C-????????????? ?????????? ?????????????????? ?? ?????????????????? ????????? (VarWar-?????????), ??? timing ?
?????????????

????????? ???????? ? ????????

2. ????????????? balance/missingness gate ??? ?????????????????? ??????????

3. ????????????? ?????? E/F ? ?????????? ?????? unsafe_demo.

????????? ?????????????? ?????????????? CUPAC ?? ?????????? ?????? ? VarWar (????????? ??????????
~34% ? ~33%; ?????? ?????????, ?? ?????????????) ? ??? ? ??? ? ?????? ?????????????? ??????????, ??
?????????

????? ?????????? B/C.

?????????????????

bash

pip install -e .[ml]

?. NOTEBOOK.IPYNB; ?????? ATE-CI ? ??????? ?????????? ?? RND_REPORTS.BENCHMARK

DELTA-???????? ?? ????????? ?????????? (??????? GITIGNORED DATA/, RAW ?? ??????????):

python tools/audit_datasets.py --delta # ? expanded_dataset_delta_results.csv + figures/

python tools/generate_pdf.py # ???????????? report.pdf